INCH-POUND
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27 February 1989
SUPERSEDING
MIL-W-8982/1
6 June 1986

MILITARY SPECIFICATION SHEET

WRENCH, TORQUE ADAPTER, SPLINED, SQUARE DRIVE, HIGH STRENGTH, THIN WALL

This specification is approved for use by all Departments and Agencies of the Department of Defense .

The complete requirements for procuring the torque adapters described herein shall consist of this document and the latest isssue of MIL-W-8982.

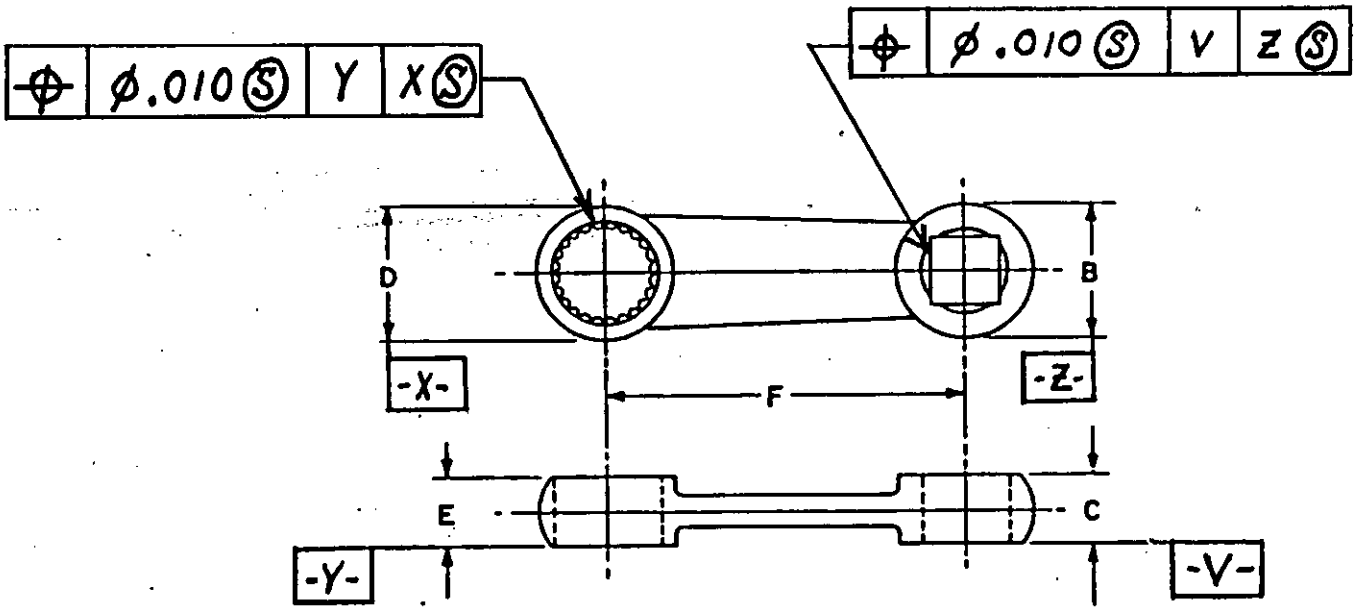


FIGURE 1. Torque Adapter.

DESIGN REQUIREMENTS:

1. The torque adapter shall conform to Figure 1 and the dimensions in Table I for 3/8 inch square drive and in Table II for 1/2 inch square drive. Spline and chamfer dimensions shall conform to MS33787. Square drive dimensions shall conform to ANSI-B107.4. For adapter retention, either the cross hole or recess shall be used, but must be specified in the part number callout. See Notes, p. 4, for example of part number. If recesses are specified, they shall be on all four (4) sides in the square drive torque adapters.

2. The torque adapter shall be hardened to 48-52 HRC.

AMSC N/A FSC 5120

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3. The torque adapter shall be chromium plated in accordance with QQ-C-320, Class 1, Type I. Cadmium plating is not permitted. Nickel underplating in accordance with QQ-N-290, Class 1, Grade G.

4. The torque adapter material shall be Grade E4340 (UNS G43406) per AMS 6415; Grade 4140 (UNS G41400) per AMS 6381; or Grade 6150 (UNS G61500) per AMS 6448.

5. Surface roughness shall be as specified in ANSI B107.7.

PERFORMANCE REQUIREMENTS:

1. The torque adapter shall withstand the endurance torque load and ultimate torque load values specified in Tables I and II. There shall be no damage or permanent deformation (set) to the wrench.

2. When engaged with the spline mandrel, the torque adapter shall be tested for conformance to the endurance torque load and ultimate torque load values specified in Tables I and II. The load shall be held for 10 seconds. Mandrel dimensions in the engaged portion shall be in accordance with MS33787, Table I, except C and W diameters shall be held to a minimum dimension (+0.002, -0.000 inches).

3. The mandrel shall be hardened to not less than 55 HRC and smoothly finished in the engagement area of wrench openings.

4. The torque adapter shall be subjected to not less than 500 torque cycles on the spline mandrels in accordance with the endurance torque load and ultimate torque load values specified in Tables I and II. Engagement length shall be as specified for the ultimate tests.

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TABLE I. Spline wrench torque adaptor dimensions and test loads 3/8 inch square drive.

Spline No.	Basic Wrench Size (in.)	B Dia of Sq End Max (in.)	C Thickness of Sq End Max (in.)	D Dia of Wr End Max (in.)	E Thickness of Wrench End Max (in.)	F Ctr to Ctr of Sq to Wr ±.04 (in.)	Spline Test		
							Engagement Length (in.)	Ult Torque (in.-lb)	Endur Torque (in.-lb)
06	0.188	0.760	0.458	0.320	0.250	2.00	0.091	160	95
07	0.218	0.760	0.458	0.359	0.250	2.00	0.110	235	140
08	0.250	0.760	0.458	0.400	0.250	2.00	0.125	300	195
09	0.281	0.760	0.458	0.447	0.250	2.00	0.140	365	220
10	0.312	0.760	0.458	0.490	0.250	2.00	0.156	450	250
12	0.375	0.760	0.458	0.590	0.343	2.00	0.188	735	515
14	0.438	0.760	0.458	0.680	0.375	2.00	0.219	1,250	875
16	0.500	0.760	0.458	0.745	0.375	2.00	0.250	2,250	1,575
18	0.562	0.760	0.458	0.825	0.406	2.00	0.281	2,500	1,750
20	0.625	0.760	0.458	0.907	0.468	2.00	0.312	2,500	1,750
22	0.688	0.760	0.458	0.996	0.516	2.00	0.344	2,500	1,750
24	0.750	0.760	0.458	1.090	0.531	2.00	0.375	2,500	1,750

TABLE II. Spline wrench torque adaptor dimensions and test loads 1/2 inch square drive.

Spline No.	Basic Wrench Size (in.)	B Dia of Sq End Max (in.)	C Thickness of Sq End Max (in.)	D Dia of Wr End Max (in.)	E Thickness of Wrench End Max (in.)	F Ctr to Ctr of Sq to Wr ±.04 (in.)	Spline Test		
							Engagement Length (in.)	Ult Torque (in.-lb)	Endur Torque (in.-lb)
18	0.562	1.055	0.524	0.825	0.406	3.000	0.281	3,000	2,275
20	0.625	1.055	0.524	0.907	0.516	3.000	0.312	3,250	2,450
22	0.687	1.055	0.524	0.996	0.516	3.000	0.344	4,250	2,975
24	0.750	1.055	0.524	1.090	0.531	3.000	0.375	4,500	3,500
26	0.812	1.055	0.524	1.170	0.547	3.000	0.406	6,300	4,410
28	0.875	1.055	0.524	1.265	0.563	3.000	0.438	7,000	4,900

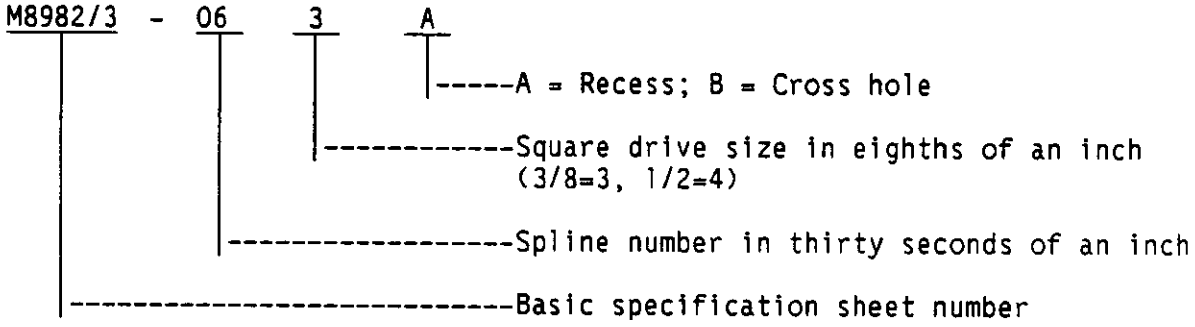
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NOTES:

1. Dimensions are in inches.

2. The torque adaptor shall be of uniform quality. It shall be free from nicks, splits, cracks, laps, seams and scale. It shall have no imperfections that will adversely affect serviceability, reliability or safety.

3. Example of a military part number:



4. For design purposes this specification sheet takes precedence over procurement documents referenced herein.

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Army - AV
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